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Linear Stages for Next Generation Precision Motion Systems

Jun Young Yoon*, Lei Zhou*, and David L. Trumper

***Abstract*— This paper presents the design, fabrication, and testing of two linear stages for next generation precision mechatronic motion systems such as in semiconductor photolithography machines. We have designed a new linear iron-core permanent-magnet motor that can simultaneously achieve high force and low noise, providing a promising candidate for actuating systems for high-throughput and high-precision servo applications. The vibro-acoustic noise mechanism of linear iron-core synchronous motors and a new magnetic design approach to mitigate such noise issue are presented in this paper, along with the experimental validation of significant higher-force and lower-noise performance as compared to a conventional linear iron-core motor. In this paper, we also present another type of novel linear stage for in-vacuum transportation tasks in precision manufacturing systems. This new linear stage system is driven by a hysteresis motor and magnetically-levitated with two-degrees-of-freedom (DOF) active bearing controls and three-DOF passive magnetic bearings. This paper discusses the design, operating principles, and experimental results of this magnetically levitated linear motor system.**

I. Introduction

In this paper we present the design, building, and testing of two linear stages for the next-generation precision motion systems: (1) a novel high-force, and low-noise linear iron-core permanent magnet linear motor, and (2) a magnetically-levitated linear stage driven by hysteresis motor with a linear bearingless slice motor configuration.

As demand for high-throughput and high-precision mechatronic systems increases, e.g., in the semiconductor lithography industry, the design and development of a new high-acceleration and high-accuracy linear stage system has become an important research topic over the years. Conventional linear iron-core synchronous motors have been a strong candidate for high-acceleration and high-throughput motion applications due to their high-force capability as compared to air-core motors. Such motors however generate significant vibration and acoustic noise which deteriorate the system accuracy performance, thereby limiting their usage in precision mechatronic systems. In order to address such issues, we have designed, constructed, and tested a new high-force and low-noise linear iron-core motor, which is presented in detail in this paper.

This paper also introduces a new hysteresis-motor-driven magnetically-levitated linear stage system designed for in-vacuum transportation tasks. A growing number of precision manufacturing applications require in-vacuum motion solutions to minimize the risk of unwanted chemical reactions or pollution of the process and surrounding equipment. The design of such in-vacuum motion systems is challenging because permanent magnets and potted motor windings need to be encapsulated to prevent out-gassing in high vacuum, which increases the system complexity. Targeting for the in-vacuum operation in precision manufacturing systems, we designed, built, and are testing a novel magnetically-levitated linear stage driven by a linear hysteresis motor. The design, operation principles, and experimental results are presented in this paper.

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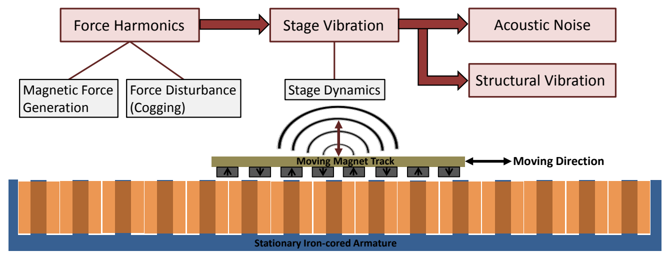

Fig. 1. Schematic of vibro-acoustic noise mechanism of linear iron-core permanent-magnet motors. Linear motor schematic shown here has a conventional design of 3-phase-and-4-magnet combination.

II. High-Force and Low-Noise Linear Iron-Core Permanent-Magnet Motor

We discuss in this section the high-force and low-noise linear iron-core synchronous motor designed and developed for high-throughput and high-precision motion systems such as semiconductor lithography machines. We first introduce the noise mechanism of linear iron-core permanent-magnet motors, and present our new motor design to reduce such motor noise. Experimental validation of the new motor is also presented in this section as compared to a conventional linear iron-core motor, in terms of noise performance and force capability.

A. Linear Iron-core Motor Noise Mechanism

When a linear iron-core motor is in operation, coil-driven magnetic force is generated to achieve required acceleration. Along with this, geometry-driven cogging forces are produced due to the relative motion between salient iron cores and the permanent magnets. These forces contain harmonics depending on the magnetic design of the motor, and such force harmonics vibrate the moving stage. Such stage vibration becomes especially severe when the harmonic

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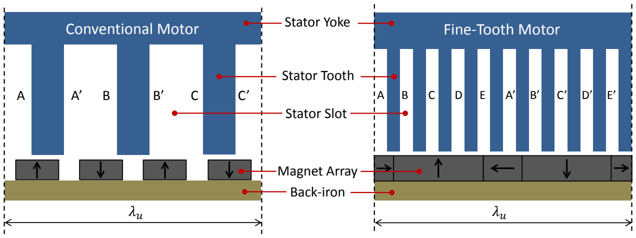


Fig. 2. Schematic comparison of linear iron-core permanent-magnet motor designs between the conventional 3-4 combination motor (left) and the fine-tooth motor (right) with a fundamental motor unit of λ_u.

frequencies align with the stage eigen-frequencies, thereby resonating the stage dynamics. Such vibration can transmit through the system structure and also radiate as acoustic noise [1], impairing the accuracy performance of precision motion systems. Fig. 1 schematically shows this vibro-acoustic noise mechanism. Note that a conventional motor design is used in the figure, but such a noise mechanism holds for linear iron-core motors in general.

B. Novel Fine-Tooth Motor Design

A conventional linear iron-core synchronous motor has three phases on three iron cores interacting with four magnets to generate thrust, as shown on the left side in Fig. 2. This is why such a conventional motor is often called a 3-4 combination motor with the fundamental motor unit length of λ_u containing one set of 3 cores and 4 magnets. Since the armature contains lumped windings for the phases indicated as A-A', B-B', and C-C' in the figure, and the magnet array is also discrete with the typical N-S-N-S pattern, the magnetic potentials produced by both armature and magnets contain significant high-frequency harmonics. Such field harmonics create force harmonics which can cause significant motor vibration and associated acoustic noise.

As a design solution to such noise issue, we use new design approaches [1] of fine iron-core teeth, narrow slots with high aspect ratio, multi-phase full-pitch windings, and Halbach magnet array, as shown on the right side in Fig. 2, so as to reduce the harmonics of magnetic fields and resultant magnetic force. We refer our new motor as a *fine-tooth* motor, and its fundamental unit length λ_u contains five phases in the armature and one pole-pair of magnets in the moving track. The tooth and slot widths are 2 mm with a slot depth of 30 mm. The stator laminations of 350 μm thickness are stacked to have a width of 52 mm, and the AWG 23 coils are wound in the slots with a total of 126 turns per slot. The vertical magnets in the Halbach array have a width of 14 mm taking 70 % of the pole pitch of 20 mm. The magnet thickness we use is 7 mm. For the detailed motor design process, see [1], [2].

C. Experimental Linear Stage Testbed

Fig. 3 shows pictures of the stator armatures and the permanent-magnet tracks of both the conventional iron-core motor (TL18 by Tecnotion) and our fine-tooth motor. The stator armature of the fine-tooth motor shown in Fig. 3a

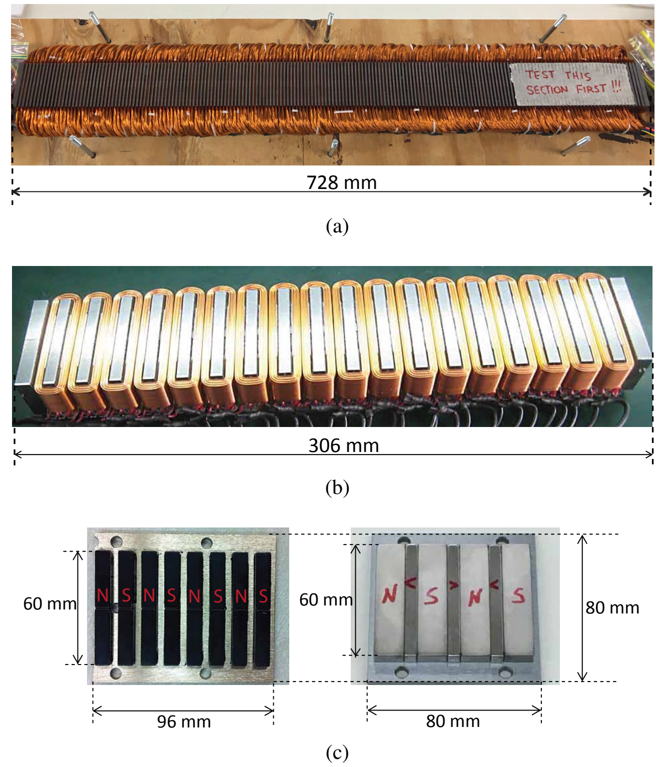


Fig. 3. Linear iron-core permanent-magnet motor components. (a) Fine-tooth motor stator. (b) Conventional motor stator. Photo courtesy of Tecnotion. (c) Permanent-magnet tracks of fine-tooth motor (right) and conventional motor (left).

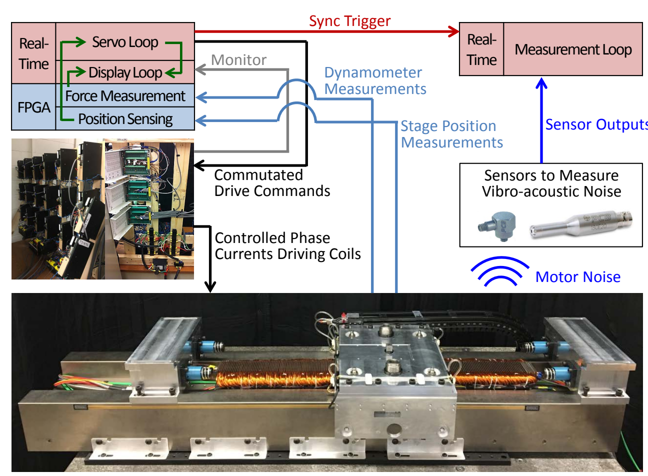
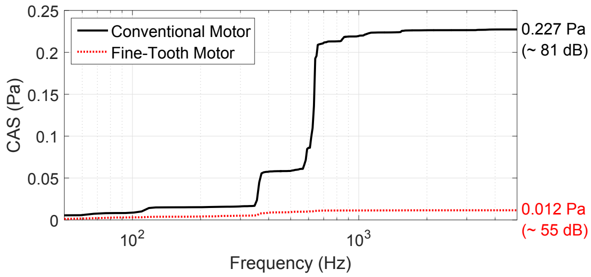


Fig. 4. Experimental testbed system with the linear stage equipped with our fine-tooth motor, power electronics to drive the motor, sensors to measure vibro-acoustic noise, and real-time controllers to deterministically run the servo, display, and measurement loops.

contains five-phase double-layered concentrated full-pitch windings while the conventional motor stator shown in Fig. 3b has three-phase lumped (or shortest-pitch) windings. As for the magnet track, the fine-tooth motor uses the Halbach array pattern with two pole pairs while the conventional motor has the typical N-S-N-S pattern with four pole pairs as compared in Fig. 3c.

These motor components are installed in the experimental



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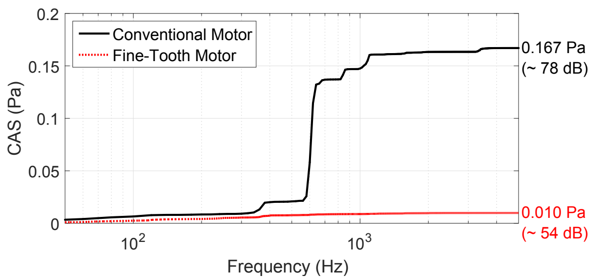
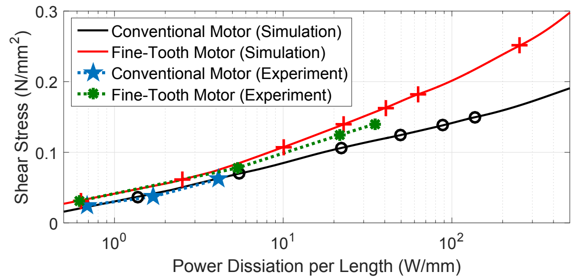


(b)

Fig. 5. Comparison of cumulative amplitude spectra (CAS) of measured acoustic noise between the conventional motor and our fine-tooth motor while cycling with a fourth-order position trajectory at the maximum acceleration of 25 m/s^2 and the maximum velocity of 1 m/s. (a) Acceleration region. (b) Constant velocity region.



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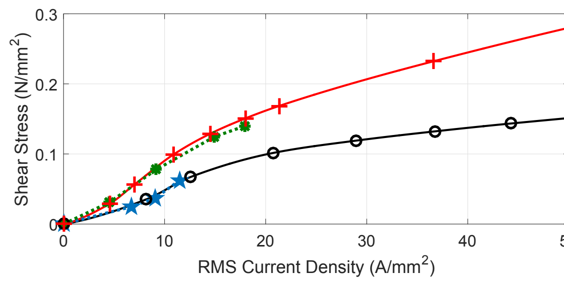


(b)

Fig. 6. Comparison of shear stress performance between the conventional motor and our fine-tooth motor both in simulation and experiments. (a) against power dissipation per motor unit length, λ_u. (b) against RMS (root-mean square) current density in winding coil.

linear stage testbed designed and constructed for noise and shear stress experiments. Fig. 4 shows a schematic overview of the testbed system. We use two separate real-time controllers (PXI-8110 and PXIe-8133 by National Instruments) to run 1) the position servo loop at 10 kHz with the controller bandwidth of 200 Hz, 2) the display loop at 3 kHz to monitor relevant control parameters, and 3) the measurement loop at 50 kHz to measure the acoustic noise using microphones (130E20 by PCB Piezotronics). The commutation algorithms and control laws are commanded by the servo loop to the power electronics to drive the motor phases. We use optical encoders (T1001 and RGSZ20 by Renishaw) with the interpolation resolution of 0.1 μm to provide a real-time position feedback to the controller. The linear stage runs on air-bearings to eliminate noise possibly caused by mechanical contacts.

D. Noise Experiment Results

The acoustic noise of both the conventional motor (TL18 by Tecnotion) and our new fine-tooth motor is measured while cycling the stage with a fourth-order position trajectory at the maximum acceleration of 25 m/s^2 and the maximum velocity of 1 m/s. The cumulative amplitude spectra (CAS) [3] of the measured noise are plotted in Fig. 5 both for the acceleration region and constant velocity region. The majority of the conventional motor noise (black solid curves) in both regions comes from the stage dynamics, especially the rigid-body mode against the air-bearing stiffness at 360 Hz and the bending mode at 630 Hz. This dynamic-driven noise stems from the conventional motor design generating high-frequency force harmonics. Within the bandwidth of 5 kHz, we observe the overall noise level of 0.227 Pa and 0.167 Pa for the acceleration and constant velocity regions, respectively, which are corresponding to the sound pressure level of 81 dB and 78 dB using a reference pressure of 20 μPa. Such significant dynamic-driven noise impairs system accuracy, limiting the usage of conventional 3-4 combination motors for high-throughput and high-precision motion systems, in spite of their high-force capability.

With our new fine-tooth motor, however, we achieve significant noise reduction in both regions. The noise CAS over a 5 kHz range is reduced by 95 % to 0.012 Pa and 94 % to 0.010 Pa during the acceleration and constant velocity regions, respectively. In terms of the sound pressure level, this corresponds to a significant reduction of 26 dB and 24 dB. Since we have less force harmonics generated using the fine-tooth design, the stage dynamics are less excited, resulting in relatively flat noise CAS curves in both regions. Such significant noise reduction performance illustrates the industrial potential of our new motor design to be utilized in high-precision positioning applications such as semiconductor photolithography machines.

E. Shear Stress Experiment Results

In addition to the low noise performance, our fine-tooth motor also shows higher force capability as compared to the conventional 3-4 combination motor. Fig. 6 compares both the simulated and experimentally obtained shear stress of our motor to the conventional motor against the power dissipation per fundamental motor unit of λ_u as in Fig. 6a and also against the RMS (root-mean square) current density in coil wire as in Fig. 6b.

As can be seen from the simulation and experimental validation in Fig. 6a, our fine-tooth motor shows higher shear stress density than the conventional motor at all power levels, especially higher force potential at higher power, as illustrated by the steeper slope. At 500 W/mm, for instance, we expect a significant shear stress increase of 62 % from 0.183 N/mm^2 to 0.296 N/mm^2. Under the same thermal conditions represented by the RMS current density in wire, we also observe higher shear stress performance of our fine-tooth motor compared to the conventional motor. At a moderate current density of 10 A/mm^2 suitable for air-cooled motors, for example, the shear stress is significantly enhanced by 90 % from 0.048 N/mm^2 to 0.091 N/mm^2. This power- and thermal-efficient high-force capability of our motor, together with the low noise performance, makes our fine-tooth motor a promising actuator candidate for next generation high-acceleration and high-accuracy motion systems.

III. Hysteresis Motor Driven and Magnetically Suspended Linear Stage for In-vacuum Operation

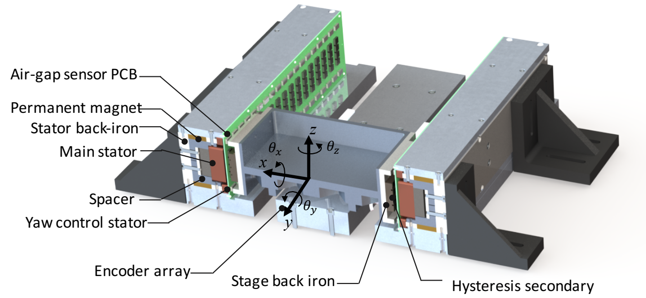
This section presents a novel magnetically-levitated linear stage driven by linear hysteresis motors, designed for the in-vacuum transportation tasks in precision manufacturing systems, such as reticle and wafer transportation in extreme ultraviolet (EUV) photolithography machines. We first introduce an overview of the prototype stage system and its operating principles, and then present the control method for the system. The experimental results of the magnetic suspension and the linear servo performance are also presented.

A. Hardware Overview

Fig. 7 shows a CAD model and photograph of our magnetically levitated linear stage system, which mainly consists of two stators, one moving stage, and a sensing system. The coordinate system we use is also shown in the figure. Here the moving stage is driven along the y-axis. The magnetic levitation of the moving stage is active in x- and θ_z-directions, and is passive in z-, θ_x-, and θ_y-directions in this linear slice motor configuration. The moving stage is designed to be vacuum-compatible. When the stage system will be operated in vacuum, a channel with thin walls (not shown in figure) needs to be configured along the motion range of the moving stage so as to separate the stage and the stators. The levitated moving stage transports the payload inside the channel in clean vacuum, while the stators are configured outside the channel in a relatively dirty vacuum environment.

Fig. 8a shows a photograph of the stator armature in a front view. The stator assembly mainly consists of one motor stator, two yaw control stators, and a flux-biasing structure comprising two rows of permanent magnets and one stator back-iron, as also shown in Fig. 7. Lumped windings are selected for the stators due to volume constraints. The windings on the motor stators are three phase, while we use 5-phase windings for the yaw control stators. There are a total of 11 phase currents independently controlled in the system, including two sets of 3-phase windings in the left and right motor stators, and one set of 5-phase windings in the yaw control stators. Current controlled switching power amplifiers (B30A40 by AMC) are used to drive currents in the windings with a DC bus voltage of 300 V.

The levitated and moving stage shown in Fig. 8b consists of an aluminum stage base, two stage back-irons, two hysteresis motor secondaries, and four bias flux collectors. The motor secondaries are made of hardened D2 tool steel since it has relatively large magnetic hysteresis. Such material also has large permeability, which is advantageous in generating



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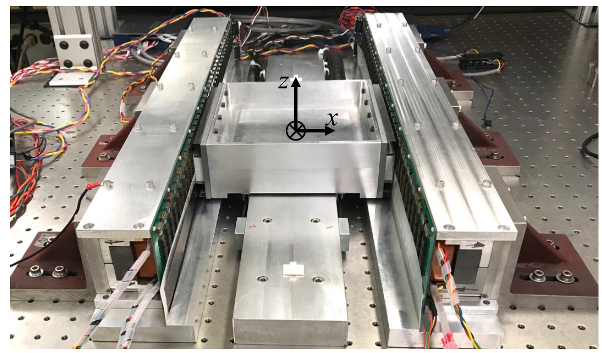
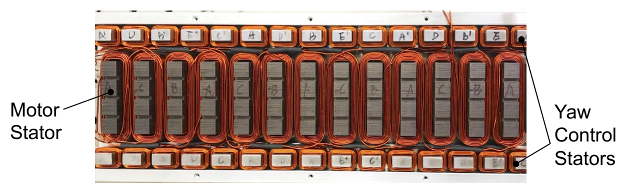


(b)

Fig. 7. System overview of the magnetically-levitated linear stage. (a) Cross-section schematic showing the key components of the stage system. (b) Photograph of the magnetically-suspended linear stage setup.



(a)

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Fig. 8. Photographs of major parts in the magnetically-levitated linear stage system. (a) Stator armature. (b) Moving stage with motor secondaries.

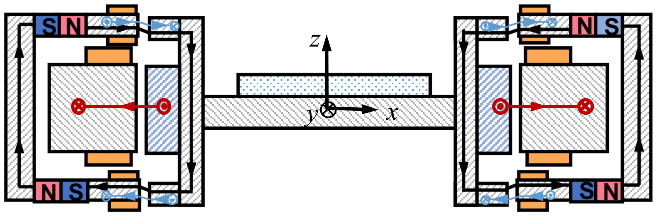


Fig. 9. Magnetic fluxes in the MLLS system. Black lines: PM fluxes for passive magnetic suspension in z, θ_x, and θ_y-DOFs; red lines: motor fluxes for x-directional suspension force and y-directional thrust force, blue line: yaw control fluxes for θ_z-directional suspension torque.

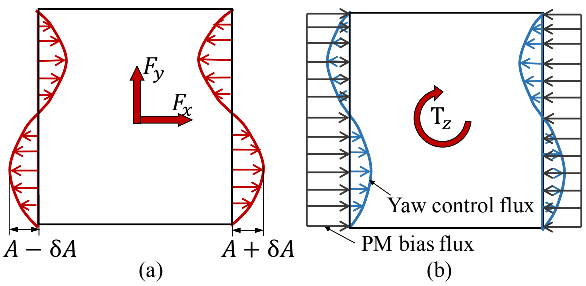


Fig. 10. Top view of air-gap magnetic flux distribution in the stage-fixed frame and means for controlling force/torque generation.

large reluctance forces for the magnetic suspension. The total mass of the stage is 4.9 kg.

We use two different kinds of displacement sensors to measure the stage motion in x-, y-, and θ_z-DOFs. To measure the x- and θ_z-directional displacements of the stage at different y-positions, a total of 16 optical displacement sensors are arranged along the stator on the two printed circuit boards (PCBs) mounted on the front surface of the stators, as shown in Fig. 7. At each position, two pairs of optical sensors on both sides differentially measure the stage motion. These displacement signals are used in the feedback levitation control to stabilize the magnetic suspension of the moving stage. In addition, two rows of magnetic encoders are arranged along the motion direction to measure the y-directional displacement of the stage with two magnetic encoder scales attached on the bottom of the moving stage and four encoder read-heads per row mounted on the base. With this sensing system design, there are no umbilical cables attached on the moving stage.

B. Operating Principle

1) Magnetic Suspension Force/Torque Generation: Fig. 9 shows a cross-section diagram of the magnetically-levitated linear stage and the magnetic flux distributions in the system, and Fig. 10 shows a top view of the air gap magnetic fluxes. There are three kinds of magnetic fluxes in the system. The black lines in Fig. 9 and Fig. 10b show the permanent magnet (PM) bias magnetic fluxes, which are used to generate passive magnetic suspension force/torque in z, θ_x, and θ_y DOFs. When the stage is displaced in these directions, the PM fluxes in the air gaps provide restoring forces to passively stabilize the magnetic suspension. Such PM bias flux however generates destabilizing force/torque in x- and θ_z-directions, and so feedback controls are required to stabilize the magnetic suspension in those DOFs. Rotary motors with this magnetic suspension design are often referred to as slice bearingless motors, and typical applications of such bearingless motors include pumps [4], high-speed fans [5], and centrifuges [6]. To our knowledge, this paper presents the first linear slice bearingless motor design to be reported in the literature. This magnetic design uses the permanent magnet bias flux to carry the weight of the stage and the payload, which does not require power consumption. Such magnetic design is attractive for magnetic suspended transportation stages for light payloads, such as reticles and silicon wafers.

The blue lines in Fig. 9 and Fig. 10b show the yaw suspension control fluxes, which are generated by the five-phase windings in the yaw control stators. In the top and bottom air gaps, the yaw suspension control flux is distributed sinusoidally, and is synchronous to the moving stage. This flux steers the PM bias flux (black lines) to generate θ_z-directional suspension torque. For example, in Fig. 10b, the yaw control flux attenuates the PM bias flux in top left and bottom right areas, while intensifying the flux in bottom left and top right areas, thereby generating a torque about the vertical direction as shown by the T_z arrow in Fig. 10.

The red lines in Fig. 9 and Fig. 10a represent the motor fluxes, which are generated by the windings in the motor stators. The common mode of the left and right motor fluxes is used to generate y-directional thrust force on the stage by interacting with the hysteresis motor secondaries. The differential between the left and right motor fluxes generates lateral-direction reluctance force, which is used to control the x-directional magnetic suspension.

2) Thrust Force Generation: Our magnetically-suspended linear stage uses linear hysteresis motors for the thrust force generation. When the motor windings are excited, the induced magnetization in the secondaries lags behind the external field due to the magnetic hysteresis of the secondary material, thereby generating thrust forces to be aligned with the external field [7]. Although hysteresis motors have relatively low thrust force generation capability compared to other motor types, it is favorable for this application since it requires no magnet on the moving stage. This allows a relatively simple stage design, since permanent magnets need to be encapsulated in vacuum to prevent out-gassing. The hysteresis motor secondaries are often pre-magnetized using a large current amplitude to improve the force generation capability of the motor. There are also additional reluctance thrust forces in linear hysteresis motors at the edges of the motor secondaries due to the end effects.

We operate our linear hysteresis motors in synchronous mode for two reasons: 1) the reluctance force in the linear hysteresis motor is oscillatory when the motor is asynchronous, which introduces undesirable vibration to the system, and 2) synchronous operation of the motor eliminates secondary hysteresis and eddy current losses. This is especially desirable for in-vacuum transportation systems because it is challenging to cool the moving stage in vacuum.

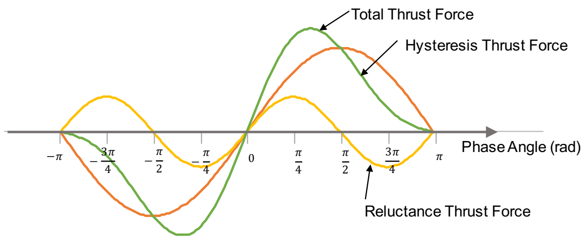


Fig. 11. Thrust force and phase angle relationships in hysteresis-reluctance hybrid motors.

TABLE I
MAGNETIC LEVITATION STIFFNESSES AND RESONANCE FREQUENCIES IN THE PASSIVE LEVITATED DOFS

DOF	Resonance Frequency	Passive Stiffness
z (Vertical)	12.9 Hz	31 N/mm
θ_x (Pitch)	10.6 Hz	76 Nm/rad
θ_y (Roll)	9.0 Hz	118 Nm/rad

Fig. 11 shows the thrust force and phase relationship of linear hysteresis-reluctance hybrid motor in synchronous operation. Here, the horizontal axis shows the phase difference between the stator excitation and the stage position. We can see that the hysteresis thrust force has its maximum values at a phase angle of $\pm\pi/2$. With the reluctance thrust force however, the peak of the total thrust force shifts toward the center. Note that this phase difference is used as a control effort in our position control loop as discussed in the following section.

C. Control Design

Fig. 12 shows a block diagram of the control system for the magnetically levitated linear stage. The x- and θ_z-directional displacements of the moving stage are estimated from the encoder and air-gap sensor measurements, and are fedback for suspension control. The x-DOF magnetic suspension control signal u_x is the differential magnitude of the left and right motor current amplitudes, and a bias current I_{bias} added to the motor stator currents for thrust force generation and maintaining magnetic suspension. The θ_z-DOF suspension control effort signal is used to determine the yaw control stator current amplitude. The position control loop for the linear stage is closed with the encoder signals being used for feedback, and the control effort signal is used to determine the phase between the motor stator excitation and the position of the linear stage. The controllers C_x, C_y, and C_{θ_z} implement standard lead-lag form of PID controllers. The signals from the D/A converters of the real-time controllers are sent to power amplifiers as current commands, which drive the motor windings in both motor stators and yaw control stators.

D. Experimental Results

1) Magnetic Suspension Test Results: Table I shows the magnetic suspension stiffnesses in the passive levitated DOFs, which are estimated by measuring the natural frequencies of these modes when the stage is levitated. Note that the resonance frequencies of the passive modes are all relatively low, which is mainly because our linear stage is operating at a large air gap of 2 mm in the bias flux path, and 1.5 mm in the motor flux path. The stage sags below the equilibrium position in the vertical direction by 0.75 mm due to its weight.

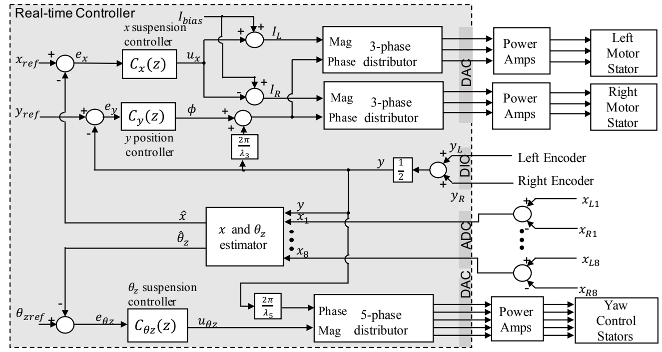
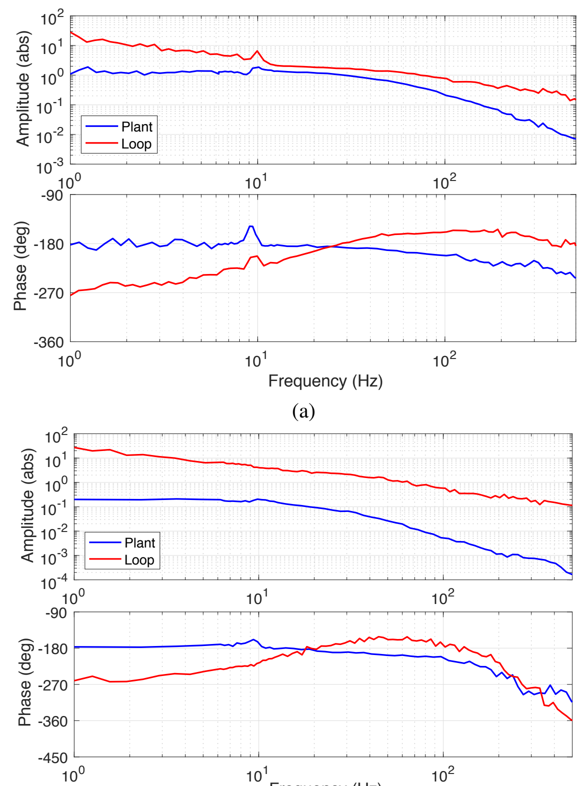


Fig. 12. Block diagram of the control for the magnetically levitated linear stage.



(a)

(b)

Fig. 13. Measured plant and loop Bode plots of the x- and θ_z-directional magnetic suspension systems. Units of the plant magnitude plots are in mm/A while the magnitudes of the loop return ratios are dimensionless. (a) x-directional levitation. (b) θ_z-directional levitation.

Fig. 13 shows the measured plant and loop frequency responses of the x- and θ_z-directional magnetic suspension of the moving stage under a bias current of 2.5 A. In the plant frequency response in the x-direction, the magnitude curves show a notch and peak around 9 Hz, which correspond to a pair of complex zeros and poles in the x-directional magnetic levitation plant. This is due to the θ_y-directional rolling mode as coupled to the measured direction. The measured loop responses show cross-over frequencies of 70 Hz and 60 Hz for the x- and θ_z-directional magnetic levitation, respectively, and the phase margins of both loops are around 20°.

2) Linear Motor Test Results: The relationship between thrust force and phase for the linear stage is measured under different bias current amplitudes, as shown in Fig. 14. In this measurement, the stage is levitated and the motor secondaries are pre-magnetized at zero air-gap with 5A current amplitude. It can be seen that the phase angles of the peak thrust force are between $\pm\pi/4$ and $\pm\pi/2$. This is a result of the combination of hysteresis force and reluctance force as discussed in Fig. 11. The maximum thrust force we achieve using our hysteresis motors is 5.3 N under 2.5 A bias current amplitude, which corresponds to an acceleration of $1100\ \mathrm{mm/s^2}$.

We conducted an initial test for the closed-loop position control of linear stage using the method shown in Fig. 12. Fig. 15 shows the measured position step response. Data show that rise time is about 14 ms, indicating that the bandwidth of the stage's position control is about 25 Hz. This test results show the feasibility of the position control for linear hysteresis motors, which we believe has not been studied in the literature before.

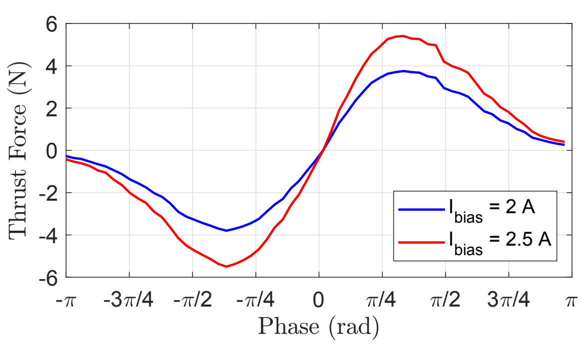


Fig. 14. Measured thrust force and phase relationship of magnetically-levitated linear stage under different bias current amplitudes.

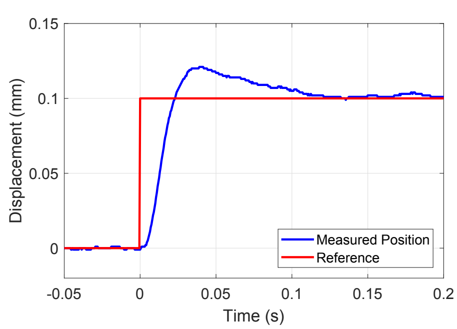


Fig. 15. Measured closed-loop position step response of our magnetically-levitated linear stage using the bias current of 2 A.

IV. Conclusion

We have presented in this paper the core technology of two different linear stage systems designed, fabricated, and tested for next generation high-precision machatronic systems. The new linear iron-core permanent-magnet motor, referred to a *fine-tooth* motor, achieved a significant motor noise reduction by more than 90 % as compared to a conventional 3-4 combination motor while simultaneously enhancing the magnetic shear stress performance at the same power and thermal levels as the conventional counterpart, for instance, 62 % increase at a power of 500 W/mm and 90 % at a current density in wire of 10 A/mm^2. Our new fine-tooth motor can be potentially utilized for next generation manufacturing, metrology, and inspection systems requiring high acceleration and high precision for high throughput.

We also introduced a novel magnetically-levitated linear stage system driven by linear hysteresis motors. The magnetic and mechanical design of such a linear stage can achieve the in-vacuum operation requirements by 1) eliminating the usage of permanent magnets for the motor secondaries to prevent out-gassing in the clean vacuum environment, 2) utilizing permanent-magnet bias flux to reduce power consumption on the active magnetic bearing control, and 3) passively stabilizing 3-DOF magnetic bearings without consuming additional power and hence minimizing the heat generation. Future work should consider the design for linear stages using the bearingless slice linear motor design, and driven by other motor principles.

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